

ActInf Livestream #032.1 ~ "Stochastic Chaos and Markov Blankets"

Livestream | 1:52:26

[Music]

hello and welcome to the active
inference lab this is actin flab live
stream number 32.1321
stochastic chaos and markov blankets
welcome to the active inference lab
we are a participatory online lab that
is learning
communicating
and practicing applied active inference
you can find us at some of the links
here on this slide
this is a recorded and an archived live
stream so please provide us with
feedback so that we can improve on our
work
all backgrounds and perspectives are
welcome here and we'll be following good
video etiquette for live streams
at this short link you can see some of
the past present and upcoming live
streams we are here in the early
november 2021
32 discussions
on stochastic chaos and markov blankets
and in the second half of november we'll
be talking about the paper thinking like
a state and we haven't yet set the
papers for 34 and 35 so if you have any
ideas or if you're an author you want to
come on to discuss
for these states
then get in touch with us
today in active stream 32.1 the goal is
to learn and discuss this extremely

interesting paper stochastic chaos and
markov blankets by fristen heinz oltofer
decosta and par from 2021

and there's a lot to cover in this paper
there's a lot of formalisms and figures
and keywords and

broader questions so i'm sure we'll have
a lot to think about and discuss
if you're watching live then please feel
free to write a question in the chat
because otherwise stephen and i are just
chilling here

so

let's get into it um with uh just intros
and warm-ups and if others join we'll
have them introduce themselves when they
do join but for now let's just um say
hello and maybe bring up just um
something that excited us about coming
to this discussion or something that we
wanted to resolve today

so

i'm daniel i'm a researcher in
california and something that i liked or
remembered about the paper was that
there was

seemingly some changes
or updates to the formalism
of core terms

and so how to understand that in the
broader trajectory of the literature
which one of these changes are like
updates where is it a software update
where is it a breaking change and then
what are the implications of those
changes for some of the broader
conclusions that people do or want to
draw from active inference and the fep

like the nature of thingness or
biological systems or levels of nested
organization
so stephen
yes uh good morning from toronto um i'm
stephen i'm
um i'm doing a practice based phd which
is kind of moves into a methods based
phd
i work a lot with
group processes and working with spatial
creative spatial approaches
to help do sense making
so i'm really interested in active
inference within that as a as a kind of
unifying way to help to
close a lot of the loops and the
fragmented
uh questions that come up when trying to
do sense making with different types of
people in different types of spaces so
um
yeah and as with daniel i'm really
interested about this paper
um as
after the initial
view of it because you see stochastic
chaos and
i'm consciously trying to not go too far
down
sort of an abstract academic aspect of
things
um
but i realized that actually these
underlying processes have got quite a
lot of applications
in this ability to understand why and
how things are unifying um

and i think that's really exciting um
because there's a lot of sort of
semi-markovian
like looks at larger cognitive processes
there's certain
practices and inactivist processes
um but what we tend to see when it gets
down into the nitty-gritty is it's it's
how
are things bubbling up through um the
kind of stochastic
chaos of of the world and life um
or should i just say stochastic
separated chaotic
processes and so the idea of having
stochastic chaos and what that means and
this paper does that
is quite useful to
start to look at what it means to have
to be able to reconcile those approaches
so
it actually does start to help think
about more applied approaches so that's
something i'm i'm interested and excited
about this paper
thank you stephen
in the
32.0
this is kind of
the lead-in was this topic of flow
and that is in a way what might start to
bridge this distinction that you raised
and will definitely return to between
kind of abstractions and applications
flow is a term that has
applications across domains from the
flow of water flow of people flow of
information and money to psychological

flow it's kind of a nomadic term it's a
transdisciplinary term
and then
that's an abstraction we're abstracting
to the patterns of flow
that help us connect the dots across
these different systems but also we want
to apply it we don't just want to say
well waters flow and information is flow
and that's the end of the story
so
how abstractions and applications
kind of work like the left and the right
foot
together to
build our knowledge and our application
almost like our epistemic and our
pragmatic value i think that'll be a
thread that's important to um
pull out what do you think about
that yeah that's a really helpful
connection
and i think this also speaks to that
usefulness of this
work beyond
modeling
directly is when when this idea of flow
is talked about in in these ways it's
kind of a bit vague right it's kind of
um and it still has this idea of things
flowing in a certain direction
um and are you in the flow and so i
think when we start to also have the
potentials for stochastic
processes and in a way the sense that
everything's flow um and you know how
does
what does low and high requirements mean

you know what does low and high
competencies preferences mean
and they kind of these are quite
bucket terms right and um essentially
they they are what they are because they
are workable
truth be told
but you know we can start to
look for something which can hold those
in a more unified way um
because i think that
um the way that we might feel in the
flow when we're in a zone
where we can both do things and we're
immersed in it
um
may may well be um
maybe what we're in all the time anyway
but it's just a different form of
um so that that also
changes you know the kind of
ramifications out there
great
point and we can actually um connect
that to this sort of view from the
outside and view from the inside which
we talk about a lot in active so it's
almost like there's the skier going down
the hill
and the view from the outside the
behavioral view is like
given the angle of the hill
they're optimally flowing that river is
physically optimally flowing down that
hill as you'd expect from the friction
and the potential energy and just the
physics of the setup
and then there's this

internal experience of flow so it's kind
of like
um maybe we are inflow all the time even
when we are getting for example
distracted or bored or frustrated
in what ways is that still flow
because if we think about flow well
flow's good you know um it's a state
where we're productive and happy and
things like that and then we're out of
flow that's what you'd hear in the
psychological literature
but let's just say that the river
we're flowing
and flooding
we don't say well we're done using flow
models because now it doesn't do what i
thought it was going to do
flow
is the whole space
so how do we think about this whole
psychological space
as being
particular regimes of flow
and flows the big topic that connects
stress
fear failing overwhelm concern
excitation etc
those are all flow states
maybe it's not the flow state that we
expect maybe not the one that we prefer
but a river can have flow states that we
don't expect or prefer either and so
it's like when we do abstract to the
physical systems we get a whole
quantitative toolkit
and we also get the ability to
maybe look beyond

phrasings that see flow as just something to be obtained and rather flow as the nature of perhaps interior experience and then we can talk about designing or perturbing regimes of flow yeah go ahead yeah that's a good point um because from the out there's the outside system like say seeing someone skiing down a hill and you could say there's different ways of doing that and then as from the inside you've got essentially a dynamic non-linear dynamical organism that's kind of trying to sort of thrive and survive and it's it's different there's no getting away from that and um the nature of that difference is that's that's another question beyond but the the the interesting thing is on that graph as well is often that graph only that the flow is only shown on the top right and not on the bottom left even though it could be seen to flow down and and um so something that's like a quadrant and it's just the top right which is seen as the flow but what's also interesting is it kind of depends again on these multi-scale dynamics okay so within certain scales of and temporal scales things start to fit our our our

organism's preferred states
um some of which is kind of built into
our
morphology
so
you know if something becomes so easy
for instance
it's a lot easier if i'm trying to
protect a building just to stand in
front of the front door as a security
guard and nothing happens all day
but
and i may end up in some sort of flow
but there's a point at which
um
there's a desire for some stimulation
there's a desire for some change
and there's also maybe
changes that happen at the kind of
this sort of ties in with with um
saffron adam saffron's work about
you know windows of experience both in
real time and over time is
when is it kind of in the same sort of
time scales that we consciously function
in
um and when is something
beyond that you know so that could be
where stress and boredom or stress and
burnout can be both because the nature
of the activity
is is too rapid it's too disjointed for
the
our morphology
uh or it may be that we perceive it that
way because we have some
you know depressive depression or
anxiety that's also not enabling us to

engage in those ways
what's your thoughts
great let's take that point to one of
the formalisms of the paper so you just
mentioned how there's like multiple
time scales and there might be slower
time scales as well as faster ones which
might even be faster than our rate of
perception or modeling
which are one in the same if perception
is indeed a generative model
and this is very early in the paper in
section 2 from dynamics to densities so
they write the aim of this section is to
get from the specification of a random
dynamical system in terms of its
equations of motion
so the
for example differential equations that
define the time evolution of a dynamical
system
two the probability density over its
states in the long term future from any
initial conditions so going from
dynamical systems which takes us into
the temporal realm
to the density realm and so there we
think about ocean water of different
physical densities
there's
a motion
to the densities because denser things
go down
and so that idea of denser things going
down or flow
from high to low density regions just
like in the atmosphere just like in the
ocean

that is going to be connected to
the behavior of
random dynamical systems
and to
information geometry so that's
a little bit of the next piece but
the two time scales are here
here's a classic flow model there's a
flow and then there's the fast
fluctuations so it's like there's the
river moving ahead at you know one
kilometer per hour
but each molecule
in that river is not always going at
exactly that speed rate
there's a fast fluctuation which might
average out to zero so in the bulk
aggregate or in the time aggregate
the flow
dominates the behavior of the river
but the fast fluctuations are also
occurring and they can often be
equivalent to a well-behaved i.e
gaussian error term
so this is where we see
the separation of time scales into sort
of the flow of the river and then the
vibration of the molecules
and those have two different
partitionings of
the system
the flow is well behaved because it is
just like a single number for the whole
river
in this simple example
and the fast fluctuations are well
behaved because they're
smaller if they were larger than the

flow they would become the flow itself
so they're smaller than the flow
and
they might be uncorrelated or have a
zero mean which allows us to do
gaussian statistics on them and then
this idea of
separation into slow and fast time
scales
is connected to density dynamics
through the focker planck equation yeah
stephen
yeah just
a little question on that so you the the
the the this time dynamic so flow
is both
at the time scale of interest i suppose
you could say
and potentially one which has a a net
kind of change in some sort of
directional kind of
gradient descent i suppose
and then
the fast fluctuations
it doesn't say random but it's probably
more stochastic um
but could also have flows going on so is
this um
is this is this um
you know how how much difference in in
the in the rate that they need to be to
be classed as
good low and fast good question
um so
the $\dot{x}$ is kind of like the
derivative of x it's the rate of change
in x it's the flow in x is going to be
decomposed

to these two terms
now we might be interested in a very
special solution to this equation
so again we have $u \cdot \dot{x}$ is kind of the
change in x which is being broken down
to these two dynamical
variables and then that's being
equivalent to the Fokker-Planck which is
more of a density framing so that's the
key
introductory step in the paper is just
to build this really strong equivalence
aka they're the same picture between
the dynamical specification and the
density specification
so if we could solve when the dynamics
were at a steady point that would be
equivalent to when the density would be
at a steady point
there might be some methods to help
solve it in one
phrasing or the other phrasing
it turns out that there's a
way to solve with a little bit more
feasibility when p of x equals zero so
that's when the flow
the density flow
given this is zero
and so that's what they're discussing
here
is u
basically
asking when does that equal zero
and that is a stationary point
so it's kind of like
doesn't mean it's just simply flat
in the reservoir but you know there's a
pump bringing water up to the reservoir

at 10 gallons per minute and then the reservoir is dumping at 10 gallons per minute

then that is going to be a point where there's a stationarity

in the overall

densities of those two bodies of water

just like if it were two bodies of heat

and so we're solving for a special

place which is where the total flow rate is zero

can i just ask a question from that without going too deep but

so

in a way the the the flows the purp the

ability of having this

relationship to the flow

is it gives it gives you a sense i

suppose you could say it's like when the

graph the the gradient zero you know

that the top of a graph between two

types of

descent and it's

it's got like that that flat gradient

um it gives you a chance to

choose moments which could have a particular

um set of properties without having to

calculate everything about it it's just

and

now when that happens what happens then

is it then that you then measure the

density and just keep measuring lots of

density points when you find the flow

points to be it

at a state of equilibrium or is there

something else so very good question

taking samples

and using that to say something about
the landscape is something we are going
to get to
welcome dean
so we will talk about how samples can be
empirically taken to say things
and do inference on the flow density
um
but you you made a good point which is
that like we want to figure out
something about a special point like
let's just say that we knew that we were
solving for a quadratic equation which
is going to come into play later so it
is going to look like a bowl
now it'd be awesome to know every single
point on the whole bowl
but it turns out we might only be
interested or mostly interested in
one point the bottom of the bowl
and for a quadratic function
the bottom of the bowl is the only time
that the derivative is zero
so we can have this big nasty quadratic
equation
and then
take the derivative of it the first
derivative
and then just ask
when does the first derivative
the slope of the curve equal zero
and we know that for a quadratic
equation there's only one point where
that happens and either it's like a bowl
or it's like an upside down bowl but
because we constrained the discussion to
a specific
family of equations which is going to be

important for variational inference
we can then
solve some very reduced
equations like the derivative which is
of a lower order of computational
complexity
and then just get the one point that we
actually wanted which is like the
location
of the bottom of the bowl
and then from the bottom of the bowl and
some of the coefficient estimates that
we have
we actually do get the whole bowl
so this is kind of like we're taking
some big family
of
equations
and then we're going to be looking for
special points like especially where the
flow is at equilibrium or stationarity
and then
fitting
a family of equations
with a form that we know we can compute
with
to
approximate
a flow that has a different underlying
dynamic and that is going to come into
play where
they show very early you can take a
lorenz attractor
which has some equations that govern its
motion so again those can be moved from
the equations of motion to the density
dynamics
and then you can reconstruct

an approximation of its dynamics
and your approximation to the dynamics
doesn't have to be even of the same form
as the generative process itself so it's
like you could fit a linear model
between height and weight
with some error
but that doesn't mean that there is a
linear model relating height and weight
it just means it's an approximation
heuristic with a form that's very easy
to solve

stephen and then welcome dean feel free
to add anything you'd like

so so in some ways this is a bit like
that particle wave challenge whereby
you know you have to use a particle to
find where something is in time and
space you have to just use a very
simplified

um understanding of the of the wave
to say okay at this point i have to i
can't

i can't have all that other information
but to have my location information
even if it's an informational location
so it's like that the flow

finding a point in the flow gives you a
way to identify
the

location maybe in the information space
and then the the foca plank
gives you a way to then
embellish that with another type of you
know put information back in
in a way

back into the into the
cake mix put a bit more

uh you know a bit more salt and a few
more uh chocolate uh
chocolate buttons in there and uh so you
can see what's going on um
okay that that's quite useful to think
about that would i be right in the sense
of like what those two equations are
doing one's
giving you
okay
some
chance to
to choose a
moment in information space or physical
space so at least you've got something
and then the other one's giving you a
way to then put some information which
you can then start to do some
variational processing on
i'll give a short thought and then dean
and blue feel free to say hello and um
just any overall thoughts and take it
wherever you want to go um
most
dynamical systems
have been framed classically
using these equations of motion so you
get like a matrix that describes the
differential equations
that describe the unfolding of the
lorenz attractor so it's framed as a
series of dynamical equations so this
early equivalence
allows us to bring some of the tools
of
fluid mechanics
and density dynamics and flow
to bear

on
these dynamical
systems
so there are some
ingredients or some
kitchen tools
that make
approximations to flow
very useful
for equations of motion that might be
intractable if you try to solve it on
just that domain
like figuring out
a chaotic system's underlying generative
process
it's basically
not possible
but potentially by approximating it
with a very specific type of
approximation families of functions that
we know have certain behaviors which
we're going to talk more about
using approximation families that are
well behaved
and
using some of the things
that have been developed from flow
decomposition like the helmholtz
decomposition
that doesn't exist for dynamical systems
so by making that equivalence between
dynamical systems and flowing systems
we can then use
flow
tools
to study
dynamical systems
as well as making it clear that we're

describing

the flow of our approximation

but it turns out that that does pretty

well at least in some of these examples

that they're going to present in the

paper

so

um dean and then blue

hi i'm dean i wasn't going to come on

today but

the conversation was stimulating enough

i figured i had to

jump in here and and

maybe add my two cents at some point

thanks

and blue

hi i'm blue i'm a research assistant in

new mexico and um

i yeah was into this paper because it uh

i don't know kind of

um piqued my interest in information

geometry and daniel i saw like in the

dot zero that you had mentioned that

course that's online i'm like gonna look

into that for sure

um

and you know i i'm curious about like a

lot of the leaps that were maybe made in

this paper between like

particles from particles to cells and

back to particles again and those kinds

of transitions get a little fuzzy for me

so i'm curious to see kind of where this

flows to

yep i think that was the information

geometry

course of john baez so that's like

again i haven't taken that but i know

that that's recommended and very cool

um

so we talked a little bit about

flow about that's kind of one of the

dreams that we could have a unified flow

model for heat and people information

behavior strategy etc

and one of the earliest moves that they

do in the paper

is just to really lock down this

connection between dynamical systems

systems through time and systems with

density so it's worth just repeating

that because that's

the second section of the paper and

that's going to be a lot of the build-up

another

term that comes into play

um is the leopardive exponent and

chaotic systems

so

if anyone has a thought of course just

raise your hand but one way that chaotic

is

quantified and formalized this doesn't

mean it's the only sense of the word

chaotic because sometimes people do use

it non-technically but the technical

definition of chaotic is referring to

the extent of the sensitivity of the

system to small changes for example in

initial conditions or in perturbations

and so there's

what's known as a leoponif exponent

which is something that can be

calculated from

the model of a dynamical system

and there are some dynamical systems

where

adjacent points

stay adjacent so it's like two pieces of wood that are just put in a river that's moving straight forward there's other systems where adjacent points converge or where adjacent points diverge and it depends where you are in the system like if you're at the bottom of the bowl then

the either way you go it's like a stable equilibrium you're going to get pushed back in

but if the bowl were upside down then either

point to the side of the top of the bowl would push you and diverge so that's the leopard exponent

and

there's

a lot of technical detail in exploring chaotic systems from this perspective so that's one matrix

um term that comes into play is using the leopardive exponents especially the leading leopardive exponents

to characterize some of the attributes of a system steven

i think it's good to mention that idea of a system you know like two things coming together and when they can meet you know the system as well they can converge or they diverge but they could also when they bump into each other there's also that

the the dynamics of the system
so as well as the of the system's
dynamics
merging the dynamics of the the change
it will stop changing once they meet
right so that there's also the extra
level of
dynamical change which you get with
these non-linear dynamical systems
which are actually working with the
dynamics of the change so it can sort of
reset itself it's not like with once
they've in a physical system they've
come together that's kind of it when
you're dealing with variations on change
it becomes much more generative
okay yes so sometimes these exquisite
sensitivities to tiny tiny differences
in initial conditions
they get swamped by random fluctuations
so
um
that's what's interesting about their
approach here is again they're
partitioning out the flow from the fast
fluctuations
and then they're describing how
when re when random fluctuations are
added in
then
the strange attractor is going to be
retraced by uh replaced by a pullback
attractor
and then the flow is the expected motion
so that's the movement of the river
of the flow
term
if this flow shows exponential

divergence of trajectories i.e
positively operative exponents we can
impute stochastic chaos
they
then
use
this approach to investigate one of the
classic
chaotic systems which has also been one
of the earliest chaotic systems of
lorenz
who
maybe there was some other livestream
where we did a little bit more on the
lorenz but it was a
meteorology question
lorenz and colleagues were studying the
weather
and how it changes and about some of the
sensitivity
of the dynamical models to tiny changes
like it seems like if it's you know 21.1
c
it seems like that should be very
similar in its behavior in the long run
to it being just 0.1 degree higher
and then as you got further away it's
like well then it'd be way more
different in the future if it's more
different today but then
it could actually turn out that a tiny
difference ends up sending your
equations haywire whereas a very
different initial condition
might actually like feed back and come
back to a similar attractor through time
so that's the kind of systems that they
were investigating in like the 1950s

and
there's been a lot of computer and
technical work since then but this is
still a total classic
and a beautiful strange attractor for
them to be working on as the first
example in their paper stephen
so what you're saying the the the
strange attractor is the kind of
complexity science idea of these
mathematical models where things get
caught in as you just described and
initial conditions
can can change the nature of what that
becomes within a certain range and
obviously outside that range it doesn't
it doesn't form so it's still kind of
off equilibrium
this pullback attractor
is a um
is is
a
almost a
a set of states which can be
moved towards
through
looking at the flows within
other types of chaotic attractors so it
is making a difference so it's
in some ways a strange attract sorry the
pullback attractor
would
or involve some sort of
inference or some sort of
some sort of driver to make it happen
it's not inherent to the to the
nature of the mathematics of that
particular

thing once you let it roll would i be
right in there
so the pull back attractor it's kind of
like let's just imagine the attractor is
like a post in the ground and there's a
bungee cord so if we were looking at
that quadratic bowl
it'd be like there's an attractor
at the bottom of the bowl and then
if all you saw was the
lateral movement of the particle so we
don't know anything about the height
necessarily or the shape of the graph
we just observe that when we push the
ball to the left
it gets pulled back as if it's being on
a bungee cord pulled left maybe it even
overshoots and it kind of dampens so
that's like a pullback attractor to a
point attractor
what's being shown here is an attractor
that is in higher dimensions so if you
had a two-dimensional pullback attractor
it'd be like the line $y = x$ and if
you get perturbed off the line you come
back down into the valley so that's like
where you start to see vector fields and
the flow of water into a river
and now
we're in a three-dimensional
attractor at least three
spatial dimensions
and so it's kind of like there's a race
car going around this butterfly track
and then it has the bungee cord so
you're getting pulled to this car
that's moving and then if you sample
your location at different times

you're being drawn to a pullback
attractor
which is the compromise between
the flow
as defined by
the equations of motions of the lorenz
attractor
and the fast stochastic fluctuations
dean and then
stephen
yeah just quickly i think this is why
sometimes using a
the the metaphor of building a bridge
is difficult because
uh not all at not all times is as
convenient as the lowest energy state in
a bowl
sometimes what we have to look at is the
fact that
at one end
call it point a
um
we're trying to figure out
what point b is and it isn't always the
lowest energy state i think here's an
example so if you ever if you've been to
hudson yards in new york
uh it's it's a fascinating translation
of a space over time and it and if you
were to ask the people who actually
constructed those
uh high trestles
at the at the point that they were built
if you did if you had asked them x
number of years later whether or not
that was going to be the lowest possible
energy state or attractor state i don't
think they could have predicted that

and so that's why i think it's difficult
at times when we
for the sake of convenience for effort
to expedite our sense of what's going to
happen next we use a connector metaphor
but in fact that's not exactly what
stochastic
possibility is about
so what we tend to do is collapse down
too quickly
down to the particulars of what purpose
a product can serve
and if we're going to build a bridge
from a product to a process which is
what i think
looking at stochastic
separation is about
we should maybe drop the idea of
a static connector and maybe look at the
back and forth and the actual dynamics
of those dynamic
situations which were now the tool
of active inference affords
cool interesting it just made me think
about like throwing a frisbee and it
catches on a roof
that's its lowest local energy state but
if you just thought about gravity maybe
you'd say well it's going to fall to the
ground so one particular trajectory
may not
realize
the ultimate lowest energy state yet
in a way it's in its own energy state
so stephen
yeah
sort of thinking again about
applications i think what dean mentions

is a good point is
there's an actual tendency to want to
have that kind of
bridge between two stable things
the thing with two stable things is it
is they're stable as we just mentioned
so therefore if you're trying to get
information about variations within them
particularly chaotic variations you're
not going to get much right so you you
kind of need
there's there's a need to be
off
out of equilibrium
to to give the conditions where things
can move in
potentially a strange attractor or
potentially other types of
you know solenoidal flows or something
there needs to be some way for flows to
happen
and then within that
which i think that might explain why
that last graph so if i'm if i'm right
in thinking that those four graphs that
you showed there
are kind of showing
that you've got this
strange
tract that you can sort of look at the
state space using
points
of where something has become moment
momentarily
zero in the the flow calculation gives
you a zero so you can therefore take a
point
and then the bottom right is basically

saying you can recapitulate that
as a global potential so you can sort of
by taking the changes
that happen in the second order you can
sort of recapitulate some kind of
understanding of the structure of the
information
is that right
there's
a lot there it
so i cannot reduce it to a binary but
yes it is showing
different aspects of how
strange attractors which are
found in these sort of exotic
deterministic chaos systems
like a perfectly clean double pendulum
and then how when we take the
approximation
we find a pullback attractor empirically
in
the approximation
so now we're describing a nicer still
stochastic but much nicer and more
approachable pull back attractor like
it's being pulled back to some
butterfly shape now we don't get these
like rings of jupiter situation from our
sampling because we're taking particular
samples and were fitting then
a flow what flow generated
those samples so we never see
rings of jupiter strange attractor we're
dealing with approximations of random
attractors
and then
we have at least two
tools slash

helpful techniques to apply
that are going to help us come to terms
with
that
approximation
so one of them is we're dealing with
flow
so we can talk about the helmholtz
decomposition so this was discussed a
lot in active 26
and as mentioned in the dot zero
stochastic chaos in markov blankets was
submitted like a few weeks after
26 was submitted bayesian mechanics
so
um
the helmholtz decomposition
describes this partitioning into the
irrotational kind of the gradient put
putting the ruler on the hill what is
the angle of the ruler if you're on the
top of the hill it's flat if you're on
the hill it's going to be at an angle so
separating that ruler on the hill
um to
the
hiking map the iso contour so let's make
the ruler flat
wherever it is whatever the slope is and
then let's just take that ruler and kind
of move it side to side as we go around
this hill
now if you're on the very top of the
hill
then the ruler is flat and there's no
iso contour
but if you're anywhere else on that
whole landscape other than a local

maxima

there's going to be some angle

to the ruler and some way to get around

the hilt

so that's the helmholtz decomposition so

that's one piece that we get to bring to

the puzzle and then

one piece that um would be interesting

to ask authors or someone with more

technical experience is

we've seen the helmholtz decomposition

before into the solenoidal and the

gradient components but then there's

this housekeeping term

um you know capital lambda

and that's described in an appendix and

it gets quite technical but it has to do

with

the

density landscape changing

as a function of flow

so it's not just like we're gonna have

one

mountain topography

and then we're gonna have rain falling

on that one mountain

there's gonna be some changes

as we flow around

so that it's more like you know you have

a jet of hot water into a cold water

you can't just take the

the mountain

of temperature

and then just ask what happens if we

pour hot water on top of that mountain

because the mountain is actually

changing the landscape is changing as a

function of the flow and some of that

gets incorporated into this housekeeping
term so one tool that we have is from
the flow toolkit is the helmholtz
decomposition

into

these different components that help us
reconstruct

a flow

that is a much better approximation

stephen

now you mentioned there that we've seen

solenoidal and gradient before with

helmholtz decomposition

when you say we've seen it before we

were saying we've seen it more recently

with the

solenoidal work

um or is the solenoid all been in there

with long

like this was drawn actually

oh yeah like we've discussed it in paper

number 26 and other papers like th this

is not a novel partitioning by the

authors

and it's very linked to other

discussions that we've had about

helmholtz decomposition in the context

of

distributional approximation

but previously was there it was more of

a focus on gradient descent

the focus was in your regime of

attention potentially

right it was

okay right going way back to early days

okay

okay so so in just without but in one in

the in that paper that was um

looking at solenoidal flow

it was it was

it was giving more weight to that wasn't

it would you say it was given more

weight was it using that in a different

explanatory way because there has been

this shift

around solenoidal flow

to give another way

beyond organicity to

explain how these dynamics can work

they're kind of co-instantiated because

they're partitionings from each other of

the same underlying model so which one

is emphasized more is totally a question

of situation and flourish

but

it also may be the case that

over the previous three years

there's more of an emphasis on

the importance of the iso contours in

complex problem solving as opposed to

just run and gun gradient descent

so

i agree with that so that's one tool we

have okay so we do the helmholtz

decomposition because again we're making

this deep equivalence between dynamical

systems and flow systems

so one tool we have from the flow world

is the helmholtz decomposition for

vector fields

now

they write

in section three one our objective is to

identify the functional form

so

could be used in the sense like what

function does it play or a functional is
like a function of a function
a functional form of the self
information or potential function that
describes the non-equilibrium steady
state density

okay so we go back to

in three

the fancy i

the fancy j

they write

the first dissipative part performs a
romanian gradient descent
on a manifold on the negative logarithm
of the steady-state density which can be
interpreted as the self-information
of any given state or as some potential
function so this is how surprise should
 i be to be in this state so now let's go
back to our bowl

you shouldn't be surprised if you're at
the bottom

you should be surprised if you're at the
top there's high potential now if it
were gravity that were pulling us down
that would be potential energy
gravitational potential energy you have
higher gravitational potential energy
that can get dissipated

free energy can be dissipated

if you have greater

potential energy

so we want something that is just like
you could have the bowl

and the ball and know where is the ball
expected to be

and where is it going to roll down from
where is there potential that's high and

where is the potential low
and then
here in figure one they're going to plot
self-information
on the bottom left that's the y-axis
self-information
and then there's that is fluctuating
through time
as the system is cranked
it's kind of like if the
self-information we're staying we're
staying at a constant value it'd be like
if you're in that race car going around
the track
then you're always where you expect to
be because again the bowl it's like a
point attractor
and then the lorenz approximation is
going to be to
a dynamical approximation
so we want to find the function
of the non-equilibrium steady state
that's going to be describing how
surprised you should be
the potential function
for that system
so these are big questions like
how is the self information measured or
calculated
and then
what does it mean for the
self-information to be high or low
and how does the self-information for
different states that we might care
about
like homeostasis
how does the self-information in those
cases differ like is the

self-information higher or lower when
homeostatically you're in a good
attractor so dean and then stephen
so i don't know this but i would guess
that on some basic level the measure is
on its most absolute terms greater or
less than
to begin with
because i think that that that relative
difference
is something that anybody doesn't even
have to be aware of consciously
calculating they can just
you're kind of you can kind of tell
but i'm not really sure but i would
think
that as as we
as we go up in terms of the the
determination and the accuracy of the
measure we'll we would we could find
ourselves
getting more precise but on its most
basic level i would think it would from
a functional standpoint
that's where we would have to begin
yes and having um
being less surprised
about the state that one is in
entails having a richer and a more
accurate generative self model
so if you have a model that just totally
off base you're going to be recurrently
surprised
if you have a very nuanced
self model
your location and your circadian rhythm
then
different observations

will be less surprising
yeah i was just feeding back what you
were saying because you did a really
good explanation
but the in in your explanation that the
the key thing was i should be more
surprised or i should be less surprised
like you you yourself in doing the
explanation come back to that basic
difference
and so again as we get more precise i
think we can we can have more
sophisticated ways of being able to show
how we arrived there but at the
beginning even a two-year-old
can can somehow tell the difference
between more and less that's why they
dig in their heels right
so
it seems like on that basic level
before we get into some of the more
sophisticated stuff
we all can it where it says addresses
these questions for one simple case i
think on its simplest terms that's where
we can probably start
we can look there first if that's not
available i don't know that the other
math
matters i know it does but to the person
who can't see that first i'm not sure
where they go
this past weekend we were hanging out
with some larva with some children and
we were really struck by how there would
be some times where it seemed like you
just would tell them something totally
different like switch topic in the

middle of a sentence and they just went along with it
and then other times where they needed something really specific like they wanted that piece of food and they want it in this part of the plate and they want to touch it this way and so it's like they have a generative model and there's certain parts that they're surprised by or not and so we're just like we want that piece of food and i'm not too surprised at the route it takes to my plate but if their generative model entails a specific way that that is going to get to their plate they're going to be surprised by anything other than that and if they don't have a generative model of topics in the world then switching topics and talking about this event and then that event it's just like it's as if any other set of topics were addressed so
great point about our intuitive connection to surprising information how that's related to our generative modeling so blue and then stephen so just um to go to your uh child model like having you know had lots of kids around for a lot of time
you know food is something particularly that they like fixate on right
i mean i uh my younger stepdaughter is like one of the easiest kids ever like she's so chill and so laid back but like i remember when she was about two years old she would like freak out like one time i cut i cut an apple for her like

sliced it up you know just to make it
like easy to eat and she like got so mad
because she wanted the whole apple the
whole thing um and she thought i was
only gonna giving her a part of it but i
think maybe food is i mean my point is
i'm rambling about the story but maybe
food is

one of these things that that is like so
baked in to our generative model because
otherwise the children have an attention
span of a gnat like you know you're in
the car and they start to argue and you
just say oh look at that pretty yellow
car right there and then they're done
they're over it like so so generally
like the you can surprise a child and
completely shift them but when it comes
to food it's like this weird and many
kids like not just you know aspen with
the apple but then there's also like
this food can't touch that food and like
they have a bigger piece of cake than me
and all this stuff so so it's food like
this fundamental like

baked in thing that that's like central
to survival and therefore it's like very
deeply ingrained in our generative model
maybe

nice agree stephen

yeah so the idea of children i'll keep
that analogy in there

if so we've got this these flows
happening and we we have

um

state spaces that get readings off so
the very first state space region i mean
unless you've got a prior somehow to go

on you don't really know what that means
and you get another one and another one
and so you get a series of state space
insights so to speak and then you'll
start to get a sense of whether
something's more or less than
i mean in the simplest term the average
right but you know we can then use
gradient descent and that to sort of
get get an understanding of that and
that with children that can be
interesting is
so yeah sometimes i just want to go with
everything and then and i notice also
like i get that with um my daughter
layla sometimes it's like you know
um
she'll she'll say like
i want something very specific or that
school there so and so went to that
school like now she knows that well i
know because that's what happened when i
was little meaning when she was a small
baby or
and it's all kind of made up
stories in some extent but it's kind of
like she's converging on she's got
enough state space information to say
okay
and and probably you've got that with
food right it's like there's there's
enough state space information to say
this and they can be prouder and i know
that and you didn't know that she said
to me you know i knew that you didn't
know that i knew that
this is quite cute but you've got that
okay so she's got enough

state space awareness
that's they're gonna converge down and
then there's other times like when you
want them to just
stay on the football pitch and not run
off to the swings uh they just go crazy
so
there's something about
you know sampling the space and saying
you've got enough information
and synchronization onto joint action
manifolds
is going to be one of the
prices of the paper
so blue as a response and then dean
so
you know super interesting about like
asking your daughter steven like you
know or the things she says that she
remembers from when she was a baby
something that's really fun to do with
kids and this also leads to like the
synchronization uh where i feel like
maybe where we're going but um
ask a kid when they can first talk like
when they can barely talk maybe like you
know two years old
um ask a kid where they were before they
were born um the answers are awesome or
like do you remember being in your
mommy's tummy like and like you know
i've heard ranges and ranges of answers
for from kids and it's really super
interesting so where is this like is it
some inherited prior is this like past
life regression or is it just like
they're falling into synchronization
they're making something up because

children have these like great
imaginations at that age but but where
does this come from like are they just
responding back to what they think you
want to hear this is like an absurd
question to ask a kid right like i mean
you can't really expect a reasonable
answer but ask because it's fun to find
out what what they think that they were
doing before they got here it's fun and
dean

first

yeah i i think it's interesting because
if you're going to go back to that
geometry

aspect of you know

can can two-year-olds think in the
abstract state yes they can

number one

and the other part i think i think where
we're sort of

we're all sort of synchronizing around
is that if it's less or more that can be
represented by the x and the y on a
graph

we've already shown that but if you want
to think whether it's in or out

you have to add that z

vector and that is the part that most
learning

doesn't necessarily explicate it's there
two-year-olds do it they know whether
they're in or out they know whether
they're getting the whole apple or a
slice

so

again i ask

are we are we attempting as people who

are trying to help the two-year-old
to build a bridge from point a to point
b
or should we be actually understanding
this as
building something that is a product
that is material
to something that is possible that's
what the stochastic field is until we
collapse it to that point b
right
and and the only way that that happens
is if we include in and out if we
include the zed
dimension to this so
again
if we
if we can't get in our own minds if we
can't get that correct
then we're actually going to get in the
way of developing pedagogy
it's i'm sorry to say that but it's
actually true so we we can think of
these little people as these wonderful
little right right brain the right brain
hasn't been brought down to
these these these equations at the
bottom of this page and yet they're
still able to function
their functional form is
self-information based on product to
process to be determined
okay so again i'm not saying this
because i've got it all picked up but
because i don't want
to have this active inference tool which
is a fantastic potential potentiator to
be collapsed to now

here's the here's the potential outcomes
that we can engineer a through zed
that that doesn't allow for a hudson
yards and yet there's a hudson yards so
if we're going to help people
let's not as you say daniel let's not
over reduce prematurely here in the
hopes of actually helping people when in
fact what that does is it says
as you fall out of the sky raindrop
you you shall depart this
this flow
uh in surrey british columbia or the
gulf of california or
at the at the mouth of the st lawrence
river we can't do that or we shouldn't
but we do and i can't understand why
the greatest thinkers do that well i can
because we like to talk
but but bottom line is we've got to
think rethink this that's what active
inference supports
totally
agreed and we can engineer the landscape
of flow you can add a dam so that the
water droplets do go one place or
another but
very interesting metaphor so stephen on
this and then we're gonna go to the next
tool that gets introduced
yeah i think
well i think i feel i know so there's a
question of it's an interesting question
as well when you say you've got a
two-year-old um
is where do they
get their knowing from so we have
abstraction in terms of thinking

abstraction

so we have well and the sort of
ontological abstractions that we can
create around our knowledge structures

but you know they have access to two
years of state space

experiential sensations of which they
can then

make a story from

whichever way they so please as they can

so there's an interesting question um

that that what it means to know that's

abstract i.e or what's outside of

the sort of lived experience level and

it's sort of and there's that that other

way and

when when one thing i always find

interesting is i see a child is around

the age of two three

you see you really see a change in the

fingers that the whole morphology of the

body shifts

and i think one of the reasons that

maybe that memories aren't accessible

from before the age of three is

literally they were encoded in a

different body that's too too far

removed

to revisit the

the priors or the likelihoods from the

state spaces that have been laid down

um so it may be partly

a natural way that we sort of forget

but we don't necessarily forget thoughts

that we had cognitively we just forget

the um the embodied stuff because we're

gonna we're gonna lay down some new

stuff unless you have a trauma in which

case that can lock in certain states
blue any comments on that
of course so you know
it's interesting um
the the
encoding in a different body i mean
we're doing that every seven years right
um like we've full cell turnover but you
know there's there's this process of
synaptic pruning that happens
um so as like we're babies you know we
grow bazillions of neurons and then
synaptic pruning happens like around the
age of two or three where a lot of those
neurons die back and then the ones that
stay uh exist and continue to to build
stronger and build better but everyone
is so completely different in terms of
like their ability to remember things
from when they were baby and not always
related to trauma like i have no
conscious memory stream until like i was
10 which is super late and like there's
no i mean maybe i might do some like
hypnosis and reveal some kind of trauma
like at some point in my life but
i mean for me to not have any like i
remember like some flashes of some stuff
i remember the photograph he showed me
the picture oh yeah i remember that do i
remember it no i probably remember the
picture of it right like so there's this
whole other level of thing but the idea
of being encoded in a different body
is super interesting because of course
my body's different now also but from
when i was 10. right but i mean also
like you know when you're 90 your body's

different from when you're 50. so what
stays what goes

how is it chosen is it this embodied
aspect of it or is it just like you know
we lock into the things that we think
are important for our model and if it
doesn't if it's not relevant for our
model maybe we just let it go
dean

i just want to say this real quick
because i know you want to move move
forward daniel but everything that we
describe here in order to get to a
concept of between
whether it's between
us today and us
seven years plus a day before that
between or any other between that we
want to describe

at some fundamental level we have to be
able to discriminate between more and
less

and what's in
and what's out

if we can do those things
we in any situation can describe between
now

once we can describe between we can see
a difference between what a product is
the bottom of this slide and a process
something that that equation
can do

if we can get if we can get that clear
then stochastic

chaos

is a between thing

that we can then manipulate to a certain
outcome or we can leave open

to possibility if we can get that
just that
we we will have conquered more than most
people have conquered in their lifetime
because
they use the word between but i don't
think they necessarily understand
what makes that up
and i think that's what we
i that's what i would like to be able to
share with others and co-learn
with others as opposed to
excuse me building another
set of specific outcomes that we can
measure and say this was you
one month ago
you a and this is you now seven years
plus a day
and now we're going to now we're going
to decide whether or not you're closer
to some thing that i've arbitrarily
selected as the thing that makes you
have learned something
if we can get there could you imagine
how amazing that would be that's just
that one thing sorry i get really
passionate about this but holy crap i'm
actually talking to people who can
actually hear that and go oh
that's interesting instead of just
dismissing it out of hand and going we
don't need to learn that because we
already know what between means
between is geometric
so for sure information geometry will
come into play and then the idea of
taking
vast

situations

and through time with uncertainty and

then calculating more or less

kafka expected free energy

that could be an extremely helpful

approach so blew anything on this or

we'll continue

yeah so what's the next tool are we

going right into the markup blanket

because man when we talk about between

oh yeah i just want to get to the markup

like it um

when we talk about between i think that

having that idea of the markov blanket

anyway i'll add it at that time so so go

on carry on absolutely the markov

blanket is um even understood by the

non-technical as a betweenness

but is it the betweenness that is your

epithelium mechanically or what kind of

betweenness are we actually talking

about

we need the tools

let's go to the next tool

so

however this is the quote in general we

cannot discount the correction term that

arises when flow operators are a

function of states

so if it were just mount everest we

could just do the solenoid and the

gradient decomposition and then figure

out how to combine iso contours with

gradient climbing and we would get to

the top however

swept under the rug by the housekeeping

term

is this

change in the underlying
mountain landscape
indeed it is this state dependency that
underwrites stochastic chaos so a way to
think about that as like the double
pendulum
the chaoticness of that system arises
from the exact angle that the double
pendulums are in you can't just say
there's a mount everest and it's the
bottom it's like okay not very helpful
for me predicting these wacky waving
shapes it's doing
this presents us with a more difficult
problem
the problem can be finessed
by using polynomial expansions
of the flow operator and the potential
as follows for n states so it could be 1
2 3 4 n dimensions
up to polynomial order m
so a linear approximation has a
polynomial order of one a quadratic
approximation has a it has a polynomial
order of two and so on so it's like
fitting higher and higher order terms so
the big idea is that we're going to use
polynomial expansions
to approximate complex underlying flow
functionals
polynomial expansions do
many things but most importantly they
restrict or scaffold our model selection
approach to a very specific form
and function and size and tractable
computation so
individuals might be familiar with a
taylor series which is where you take a

given point and you say well what is this function at that point like the sine of zero is zero and then you say what's the first derivative that's this red line and then you say what's the second derivative okay it's starting to curve a little down okay what's the third derivative and so on and you can use the taylor series to approximate that function

further and further out from your entry point that you're calculating the first second third fourth fifth derivative on etc

and then

this is a static

approximation so this is something that might be learned in high school or college level maths

a topic that's very related but not always discussed is the volterra series and so

the volterra series is a model for non-linear behavior similar to the taylor series it differs from the taylor series in its ability to capture memory effects the taylor series can be used um for approximating the response of a nonlinear system to a given input if the output of the system strictly depends on the input at the given time

and this

is developed quite extensively the volterra in

spm textbook by fristen at all so again many of the seeds and roots of what we're talking about today

check the spm

textbook and documentation

because that's where a huge amount of
this

competency in these authors arose from
and how they represented it almost 15
years ago

so

what happens when we do this polynomial
expansion of the functional form well

first off we moved the problem that we
were solving from static mountains only
to

trampolines that are moving as we're
walking

that's what brings

state-dependent changes into the mix and
action as well

and it moves us from again a static
landscape that we can just calculate
where the rulers are flat on

and then just figure out the best way to
get there well you can't just lay a flat
ruler if they're state dependent changes
because where the ruler's flat at $t = 0$
is not where the ruler is flat literally
the next moment dear

yeah i think this is excellent daniel

because what this now says is

now we can explain why there are
impressions

why there's some residue left over why
the in and out aspect of this not just
the bump

but the in and out going through that
bump

also has to be

accommodated or exported at least

acknowledged as part of everything that

we're looking at here the whole thing
so yes
that's a great explanation thank you
and the learning landscape isn't just
the mountains the interaction landscape
is more like it's like the coupled
systems we'll get to soon but it's even
one person learning landscape
is like
being in trampoline world sometimes
so
that brings us into this
space with many important topics that
are discussed
in an earlier
and different way by the spm textbook
now how good
is or could be this polynomial expansion
they write inspection of the lorenz
system suggests that a second order
polynomial approximation is sufficient
given the flow is second order in the
states
so
we're fitting an approximation that has
two terms so it's like a quadratic
so that's why we talked so much about
the bowl
because the approximation that we're
fitting
is bowl-like
which allows optimization techniques
that are simple like convex optimization
those are easy roll the ball to the
bottom of the hill by minimizing the
potential function
which potential function the free energy
potential function the free energy

functional

whereas if it was a rugged optimization

landscape

you get the whole headache of how do you

know when to stop and what if there's

just a whole neighborhood you didn't

even go to so

it seems like

potentially a second order approximation

is sufficient

then they use this word onsets which is

nice it's an assumption about the form

of an unknown function

made in order to facilitate solution of

an equation or other problem

so now we're going to go with this idea

that if we can

have a second order approximation to the

lorenz we're going to be doing good

enough let's work with that and see if

that turns out to be the case

so it turns out

if this thing that's so hard to model

the self-information the surprise

the potential function is modeled by a

second order polynomial

then the non-equilibrium steady state is

approximately gaussian

this is known as the laplace

approximation in statistics which is

also in spm very helpfully

here we generalize the notion of laplace

approximation to cover not just the

quadratic form of the log density but

also the solenoidal flow that

underwrites non-equilibrium dynamics

so the laplace approximation in

classical statistics

is like we're going to take the density distribution of what we're trying to estimate and we're going to find the peak the maximum point and then we're going to fit a gaussian over that peak and so if there's like a big shoulder and then a tiny shoulder you're just not going to get that tiny shoulder right but that's a very rich distribution if it really has that structure in the real world what you're going to get for that distribution is a gaussian centered around the big peak so for some distributions you're going to be way off there are things that are hard to estimate and there's things that don't work well with the laplace approximation it's not a solution it's an approximation however if we're interested in the mean and the variance around potentially the bulk of a distribution then solving the parameters for the laplace approximation is as easy as fitting a quadratic which is to say the ball sliding to the bottom of the bowl and once you go beyond second order approximation you have a third order quadratic there's a whole another world because then the ball could get caught in that first little elbow or it could fall off the other side

so higher order approximations
they take more computational resources
and while they might have increased
accuracy
they can often be less tractable to
actually solve so this is
at least
in my reading a pretty fundamental
contribution to
the statistics of complex systems
because they've taken the notion of
laplace approximation
not just towards estimating the mean and
the variance of some distribution out
there
but
to
the solenoidal component of flow past
helmholtz decomposition related to
non-equilibrium dynamics
stephen so
like at up to this point we actually
don't need markov blankets yet it's
giving this we haven't used the word so
how's not going to play has not come to
play yet at all
exactly exactly
so
it's telling so you've told the story
of what needs to be in place for
there to be the the kind of
state spaces
and then the nature of these state
spaces being so changeable
and non-linear
and then
there's this on top of that then there's
a way to

attractively work within that which is
the Laplace approximation in statistics
which gives you
does it always give one Gaussian on
any landscape or is there multiple or is
it just one per landscape is that yes
per
dimension per state the Laplace returns
one and only one mean and variance
it's like fitting a Gaussian
distribution to the log density
so it does only give one which is why I
gave the example of the two hump it's
going to pick one
and go there and all right so yeah let's
look at figure two and so
how interesting though that
the papers mark off blankets and still
cast the chaos
but we're not even at the Marco blanket
but the way that it gets led into
is very
um
improved and different than what we've
seen before
instead of assume a partition
dot dot dot dot
we're starting way back so now
here on the left
is the flow of the Lorenz system
using the Helmholtz decomposition
so here are the three components of the
flow as they've described it the blue is
the gradient
the orange is the solenoidal flow and
then the gold is the correction
housekeeping term
right

this uses the same format but for a laplacian system based upon the lorenz in the upper panel so we're going to be fitting a laplace like approximation to each of the states each of the dimensions of this lorenz strange attractor the key difference is that the dissipative part of the flow operator and the hessian are positive definite so just really quick matrix vocabulary the jacobian matrix is the matrix of the first partial derivatives so in the x y and z dimensions or however many there are what is the angle of the ruler that's the first partial derivative the hessian is the second order partial derivatives so it's describing the local curvature so if someone said my ruler's flat and it's curving down you're on the top of a hill if your ruler's flat and it's curving up you're at the bottom of a hill and so on so that's how the zeroth the value of the function the first derivative and the second curvature are used like in calculus so the hessian is positive definite which means the gradient flows converge to the maximum of the non-equilibrium steady-state density this is reflected by the blue arrows that point to the center of the state space

so in the lorenz world
you're getting race card around the flow
is taking you all over the place yet
by making the laplacian approximation
all of a sudden the flow
the attractive part that's dominating
the movement within that space it's
actually convergent
so that's not to say the lorenz system
converges to a stable point
it's to say our approximation parameters
converge towards potentially a more
tractable and simple attractor
in the abstract approximation space
and we'll see how well of an
approximation that indeed is but that's
the key difference is here the blue
arrows are taking you on a wild ride
as the gradient takes you around the
lorenz flow here we're doing a flow
composition on our approximation which
we constructed to be of a certain type
and it turns out to have properties
in its attractor that are very different
than the loretz
dean and then stephen
uh unmute dean and then yes
here's a question for everybody here if
if we're talking about
a
process could we be looking at those
over time seeing those arrows in both
the lorenz and the
place decomposition
thickening or expanding or
like having water added to them is has
anybody contemplated whether in fact
that is what is happening over time

so the fit is actually from a within
that the space is getting filled up by
these arrows growing
in terms of their circumference
very interesting question
the length of the arrow usually
describes the amplitude like the speed
of the flow in that moment i do not know
if i've seen a visualization where the
tree rings the arrows represent thicker
or thinner vectors but great question
and if anybody with more experience in
vector math
has a thought that could be very
interesting to know okay
stephen and then blue
and just to clarify that i've got this
right but in each of those um
points where you've got the blue arrow
and the solenoidal flows so each one
would be
solved so to speak for that point as a
as a um
you know a gaussian
approximation and then by this is like
an assemblage of multiple
results of a gaussian approximation and
it's sort of mapping them out so to
speak and so you know this this idea of
multiple
pieces of information and nestedness is
it sort of comes into play with this
yeah there's some landscape
and we go out there with our ruler our
jacobian and our curvature ruler our
hessian and then we're going to make
just transects on the hill and we're
going to just sample a bunch of points

and add a bunch of points you know one
comma one one comma two
we're going to use the ruler and the
curvature
and we're going to look at at that point
where the field is
pointing
and then
if you've sampled at um in inadequate
resolution you're not going to see your
coherent
vector field
so you need to sample at a resolution
that allows the vector field to be kind
of obvious
so blue
so i just want to point out for people
who might not know that
you need to have a positive definite
hessian
to calculate a covariance matrix like in
statistical analyses that i've done this
is like one critical
point and if you don't have an
invertible hessian then you have to
recalculate your model because
all of your factors are too entangled to
find out how they are covaried together
or separately so i just wanted to for
anybody who doesn't have the background
in in matrix math um
point that out
yes thanks dean
so blue help me does that mean literally
that you can't thin
one of those
arrows
you can't you can't let certain things

be purged from your state space
because that's like i really don't know
i'm not questioning i'm asking a
question from a statistical standpoint
and then being able to translate that to
the physical
reality
so i don't translate to physical reality
well my brain just like doesn't work
that way
but but what has to happen is what that
really means is you can't establish
conditional independence
without having some idea of the
covariance and so if you can't calculate
the covariance you can't discover this
conditional independence which is where
we're building up to
in
in specifying the markov blanket so i
just want that's that's why i wanted to
just point that out because you have to
have that
ability to find that that covariance
okay let's go to where the covariance
comes into play and then stephen so
we're gonna address
how well does this approximation work
because i mean we're fitting a few
quadratics
to like the og complex system is that
really going to work
what's being shown in figure three
is
um on the very top so this is the left
one is second versus third first versus
third and first versus second so these
are like three projections on a cube so

we're looking on a flat screen or piece
of paper about a three-dimensional
state space
and the red dots are the sampled
trajectory
in that cube so it's like there's a
firefly moving around in the cube and
we're looking at it this way and so
we're projecting it onto that
screen
and then in the background
we see this very nice
unimodal
gaussian
attractor
so the lighter part is like the
attractor part and then the darker part
is
higher potential function
more surprising less likely
so that is the lorenz trajectory
being played out
in our
laplace-like approximation cube
and so it can be shown that the flow of
the laplace approximation
it has some of that kind of butterfly
race car dynamics
it's probably doing a lot better than
just sticking a linear equation through
it but it's not perfectly recapitulating
the underlying lorenz system so how good
is the approximation
well
one thing that this approximation is
going to live or die by is whether it
can characterize
the lorenz system as chaotic or not

because if it turns out that the way
we're going to deal with chaos systems
with a positively open of exponent is
fit a non-chaotic model oh i made an
oscillator pendulum to describe
homeostasis

okay so you made a simple model great
but what's really interesting is if our
approximation

could also reflect chaos yet handle it
tractably so what they do in formalism

12

is they show that the estimated lyapunov
dimension

for the approximation

is

2.48 and they use some um numerical
approximation techniques here like by
sampling a grid

kind of that transect sampling idea but
for the lyapunov exponent at those
points rather than say just the gradient
and that favors

um or that um compares very favorably

2.48 2.43

importantly they're both like similar
and way over one

with the actual analytical

lyapunov dimension

so

the approximation is actually
highlighting the chaotic nature and
almost quantitatively the extent of the
chaos because there's more and less
chaotic systems

and so

for our purposes the laplace

approximation is easier to handle than

the lorenz system because the functional forms of the flow and potential are immediately at hand

that's the whole point we used an approximation family of equations that's going to be like having a ball rolling down to the bottom of a bowl rather than being swept up in turbulent flow

figure four is where we get to the covariance that blue mentioned

okay so we're going to have three states those are the x y and z states or dimensions of the lorenz system

and now

um we can look at the log of so the log is just sort of a transformation to help highlight some of the differences between zero and one especially

we can see the log of the jacobian which is that first partial derivative the log of the hessian which is the second partial derivative and then the log of the covariance

in the middle which is like what blue was referencing so here

the first and the second states dimensions

have very high correlations

neither of them are correlated with the third so the on diagonals are states being correlated with themselves the first and the second dimensions have high correlation well what does that look like when we use that cube shining light so on the bottom left here

we can see like the first and the second

have a like an oblong relationship
that scatter plot has
a significant regression through it
there's a correlation coefficient
there's mutual information there
there's non-sphericity
whereas the other ones that are
uncorrelated relationships have what
looks to be more of a spherical
correlation like if you sample variable
a and b and it's just a circle
there's no mutual information
and so interestingly
the covariance
matrix
looks a lot like the hessian matrix
so
that is providing evidence
that if what we care about is the
covariance amongst the states
we're getting a very similar structure
when we use
the second order polynomial
approximation
to those states
and it turns out that it does
pretty well
if you want to for example estimate
the first state given the second so
what if we could only measure two
dimensions well we wouldn't want to pick
one and two because they're basically
correlated in their measurements
we would want to pick two and three and
then do a really good job at predicting
one given two
which is to say that the informational
dimension of this system is

between two and three
it's two and a half basically
so those are some of the connections
between covariance amongst different
dimensions
and
polynomial expansions
to
dynamical systems based upon flow
approximation
and then just to summarize here then we
can have any questions and then in the
last 20 minutes we'll look towards the
markov blanket we still haven't even
gotten to the markov blanket yet
so just this is the um the foothills of
mount markov
and
they write it's important not to compl
conflate
it's important not to conflate the
simplicity of a non-equilibrium steady
state density
that's this one
with the complexity of the underlying
density dynamics
the map is not the territory you just
cannot say it enough simple
approximations are not claims about
system simplicity
in other words when prepared or observed
in initial estate the probability
density can evolve in a complex and
itinerant fashion on various sub
manifolds of the pullback tractor and
they're going to use information length
to talk about that but the important
thing to know is just that our

approximation has attributes that resonate with the original underlying systems dynamics as well as attributes that we specifically know are different or they might be different or not but no matter what they're not claims about the underlying system so that's not a pipe
stephen

yeah i think this also speaks to this idea of triangulation or multi modality

is like say this this isn't directly speaking to what's out there or what the system or or what you could say what whatever the things are that have been inferred

so there needs to be or that the more approximations that can be triangulated as i say which is kind of what happens in qualitative research qualitative research is all about the triangulation of research not about the exact measures and accuracy within research

then you can

get a better

grip

potentially on what might be out there if that makes sense so like you say with that pipe you kind of

you know if i could if i could go in and you know i've only got i've only got one aspect of that reality so i can only resolve it to a certain degree um and it is what it is and then i i'd have to see i i don't know how big that is as a painting how you know until i see it in a gallery etc etc right so

um

yeah i think that that this this

um

the need to find the way to get an

approximate weigh-in

means that you need to have many

approximate ways in to

start to make

a higher level confidence in what you're

thinking

thanks stephen dean

yeah daddy can you go back slide 48 for

a second yes

so this is the fascinating thing about

contrast and proportion

because you've got nine

squares up in the upper right there with

the jacobian and the

and the covariance and the hessian

trying to translate that and that then i

understand this is this is the big uh

this is the sort of the the

grail of translation to translate that

and the

contrast and the proportion that you see

there to fresh eyes to new eyes they

might look at that middle one

with it with a completely different

sense of what that pattern is describing

for us than the than the two outside

slices of bread

to us because we're a little bit more

familiar with it the contrast in the

proportion you can actually see where

they has cn matters from a statistical

standpoint but drop that into a context

now where you've got mountains and

rivers and lakes and forests and streams

and how do we take that information
there which is which is really quite
clear to people who are familiar with it
and translate it to the context the
situation in which we find ourselves and
are now trying trying processing to a
converged sense of context what does
this
what does this envelope tell us right
we're inside of this cube now how do we
get outside of it use this tool
to so we to re-enter
the context and have a a
an upon sense of what covariation
means
this is this is the grail if we can set
up our learning experience
as a situated one where we take this
tool in
but don't confuse the tool as you say
don't confuse the math
with with the field
that's that's again
it's it's fantastic if you can get a
whole bunch of people seeing
what what's the big picture here what's
the tool
what's the contrast in proportion with a
tool
and then what is the context what's all
of this fractalized stuff out there in
different colors
and different shapes and different sizes
how do those how do we put those two
things together into some sort of
coherent narrative
great point
so in the last

10 to 13 minutes

we're going to finally converge to

the m word

markov blanket

um and then next week it would be

awesome to speak with any of the authors

or anyone else who would like to share

their perspective

or

we'll continue because like

we're basically halfway through and we

haven't gotten to the core piece of the

markov blanket so we're gonna just look

where we're going

in the last little bit of dot one and

then in dot two we're gonna like you

know land on that markov trampoline and

then see where we go

okay

in section four

they're going to repeat the analyses of

the previous section

approximating two lorenz systems coupled

to each other through their respective

first states

this induces a richer conditional

independent structure from which one can

identify internal and external states

that are independent when conditioned

upon blanket states so

this is

the more classical markov blanket

definition which is blanket states are

those

that conditioned upon them make internal

and external states conditionally

independent so that's a definition that

we've seen before and the internal

external and blanket states they all are
co-instantiated it doesn't make sense to
talk about one without the other
and
we've mentioned that before so in
um
figure seven it's a lot like figure
three
where we were looking at those three
projections of the cube of one lorenz
system well now we're looking at two and
it's pretty clear from looking at their
trajectories
as well as just from their movement in
the state space like these are pretty
synchronized so
some
coupling has been instantiated
no different than if you had like two
engines and then you connect them with a
heat transfer device they're gonna have
coupled temperature fluctuations except
now it's like more statistical
figure eight
is a lot like figure four so in figure
four we had this three by three
matrix describing the first partial
derivatives second partial derivatives
or the covariances
of states one two and three and the on
diagonals were staged with themselves
and the off diagonals were states with
other states
now
we're going to have
three plus three states
two sets of three two coupled lorenz's
one two and three is the first lorenz

and then four five six are the three
states of the second lorenz
you see that in the jacobian there's
high correlation
within
each lorenz system and then there's that
one and four
where there's a strong value that's the
coupling like that is the cell that
represents the coupling between the
first upper quadrant and then the lower
right quadrant
and that induces some extremely
interesting covariance structures
that is what we want to investigate it
turns out that by looking at the hessian
of this system
some partitions fall out
those partitions
can be numerically
analyzed to show that
[Music]
basically there's
partial correlations
that approximate the hessian of the
lorenz systems that are coupled
so
we can use
the hessian which we already established
the second order polynomial expansion is
like basically adequate
we can use the hessian of our
approximation not of the system not the
system has a hessian our model of the
hessian
approximation
very well
aligns with the partial correlation

and so that's demonstrated
by showing that two states that
shouldn't have correlation third and
sixth and fourth and fifth so like here
three and six are not correlated
we can show that even if they start very
correlated like they just happen to
start at a similar value
the dynamics of the system actually move
them towards having a zero or even a
slightly negative partial correlation so
our

hessian approximation
is giving us deep insight into the
partial correlations
and then i

think a great place to sort of um

[Music]

posit

is with this connection of the graphical
bayesian network that we've looked at
before

with that same partitioning into four
states internal external and blanket
states now fristen and others

then partitioned the markov slash pearl
blanket states

which are of one kind

into these two kinds of blanket states

incoming sensory states and outgoing

action states that's what links us with

perception cognition action and control

theory and

planning as inference so this is an

image we've seen before

and here are those six states

so one two and three that's lorenz

system one one then four five six is

lorenz system two
and one and four is the only cross
coupling that we're putting into the
model that's this cell right here so
we're cross coupling one and four you
know you have two neurons
they have two neurons and then there's a
little tin can between the two of you
and that's inducing
a global structure to all six nodes so
you don't need to have every neuron
wired to every neuron
there's sparsity in this connection
structure
and
technical details aside
that is going to relate to the
partitioning which we're going to be
deriving
a partitioning
on the hessian
which is a second order polynomial
expansion
of our
model
so
i think um
that is um a great place to
go for today
and uh just then just to close and we
can have last thoughts or what we want
to talk about next week just to
reiterate there's no claim either the
original lorenz system or
coupled lorenz system possesses a markov
blanket so the territory doesn't have a
markov blanket our coupling doesn't even
have a markov blanket

the claim here is that there exists a
laplace approximation to these kinds of
systems

that in a virtue of the zero elements of
the hessian feature markov blankets
so not has a markup like it is wants to
be did have

those are
speculative interpretations that maybe
future larva will discover to be the
case or not

but what this paper actually claims is
very far from that

blue and then stephen

okay so here we're going to stop just
since we're getting good just as it's
getting really good okay um it's the
cliffhanger for next week i just wanted

to leave off with where this figure is
in the paper like right under that
figure it says the names active and
sensory or blanket states inherit from
the literature where these states are
often associated with biotic systems
that act on and sense their external
mulu

so

i'm like why are they imputing sensory
states onto particles like this is where
the great leap in the paper kind of
happens for me and i'm like okay now my
particle

is sensing

which is like to me okay like i'm
actually okay with that like you guys
know you've seen me like

i'm pretty good with pan psychism at all
at overall and um it doesn't really make

a big deal

big difference for me um i'm still very

interested but

it's something that a lot of people i

think are going to get maybe uh tripped

up by so i would like to hash that out

and to say well there's absolutely no

claim that real systems have a markov

blanket

you know like the skin

it's like

are we going to be talking about real

systems and how it maps onto physical

components of the real world and the

mechanisms of sensory transduction

or is this about the hessian

approximation

to the laplace

so

sometimes very close to each other in

the text

so that can make it a little bit

ambiguous stephen

yeah

um very very cool yeah blue blue makes a

good point

i suppose what i'm sensing here here and

correct me if i'm wrong is that this is

showing a way to show that patterns can

be existing

within the covariance of um

attractor states or attractors

just in themselves it could just be a

blob of stuff right in amongst blobs of

things and stuff and particles and then

it also shows that

there can be

the ability for

those
dynamics
to partition out
so it's
kind of
gives a bit of two ways it gives the
partition outpieced
and then it also gives
a way to have a blob
this is a very technical term a blob of
very
useful stochastic chaotic states which
don't need to be themselves
blanketed
but could be what becomes just sort of
sensory states for instance
but could be a reservoir
within a bigger reservoir and would that
be
a correct
sort of assumption if we truly had no
difference between our approximation
and our incoming observations like we
had a perfect model
then
maybe there'd be a valid claim that the
map was the territory and for systems
that we define in software it actually
can be that way
but for any real system this
approximation there's always going to be
a delta between the incoming
observations and our approximation and
so the claim here is that it's the
hessian the second level polynomial
expansion of our approximation that has
some very nice mathematical properties
not that the system has that or any

specific properties

dean

yeah i would just

speak to the idea that the skin is a is

a great example because

you can look at it only as a product as

a limit

or you can look at skin as a process i

mean we know that

all the processing that's going on there

so

again

do we do we do we collapse it down to

one type of function as a product as a

limit

or do we see it actually as a process as

well as as a something that's dynamical

something that's constantly changing and

then that opens up the idea of what

between means now i don't know if it

takes it down to the particle level

seems like it does at those at those

blank scales but i'm not really worried

about that so much as i'm i'm

interested in the idea

of what

betweenness is

because i don't know that skin

automatically

is a partition sometimes it's a process

as well so if we keep that open that'd

be something interesting to take up next

week

great yeah i think if anyone else has

any

comments like what a fun

paper and dot one we

hit a lot of the warm-up

stretching and preliminary topics
and contextualization that brought us
like right to the vista
of mount markov with further clouds
behind even that
but we're where we know that we can step
into the dot 2
with a backpack that's prepared and then
these are really interesting and clearly
either unresolved and or uncommunicated
and or unapplied questions in
literature in the state of the art which
is what is the relationship between
these formalisms and specific
nouns and verbs
in the world the kinds of things that
we're semantically engaging with and if
it turns out that the semantics of a
markov blanket are not like the
semantics
of any other thing
or they're importantly different from
the semantics of
what we currently feel familiar with
great
or it would be interesting if they do
map or overlap or encompass or nest
within some other semantics that already
exist but that's the question
so um
we can just look at our final
closing questions and if anyone has a
thought otherwise 32.2
next week on november 16th
and
we hope to hear from you if you would
like to
join the discussion live just get in

touch with us or contact us at
activeinference gmail.com
or just show up and
ask questions or make comments in the
live chat
but
interesting stuff
we were
a little trepidatious with such a
complex paper
but
i think bringing simplicity
and simple rules underlying complex
behavior and how can our approximations
be tractable even when the world is just
infinitely surprising
those are some of the fundamental
questions that this paper and active
inference address so blue steven and
dean
great work
and we will see everybody
next week
bye